# SDC Hackathon

# Technical Tools Declaration

## Hosted by Xiamen University Malaysia (XMUM) AI Club

Deadline: 25 October 2026, 07:59:59 MYT

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

# 1. TEAM AND PROJECT OVERVIEW

## Team Name

Rising Rookies

## Project / Solution Name

PathFinder

## Name (Leader)

LIN ZIRUI

## Email & Phone (Leader)

1094184601@qq.com | Phone: PENDING

## University Affiliation(s)

Xiamen University Malaysia

## Live Prototype URL

https://futurepath-workforce-radar.spryhaven13.chatgpt.site

## Code Repository URL (public / access granted to judges)

PENDING team public GitHub repository URL

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.1. System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

A static browser UI accepts synthetic persona selections and current skills. It fetches public title/tag/date/link metadata directly from Arbeitnow and HN Search by Algolia, validates records and removes duplicate source URLs. Jobs and AI-discussion evidence remain separate. A deterministic keyword recommender computes scenario-weighted role scores, gaps and introductory study steps from a curated catalogue. Users choose the target role and mark progress. The browser stores selections, signals and progress in localStorage until reset; JSON export is optional. ChatGPT Sites hosts static assets through Cloudflare. No profile backend, application database, runtime LLM, trained forecast model or GPU endpoint exists. Python standard-library serving and a .replit run configuration are included for team deployment, which has not been performed or evidenced. No profile data is sent with feed requests. Manual refresh triggers recalculation; no background monitor runs.

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.2. Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name/version | Purpose | Licence |
|---|---|---|
| HTML5; CSS3; JavaScript ES2020+ | Runtime interface and deterministic recommendation logic | Web standards; original app code under Apache 2.0, team IP review pending |
| Browser Fetch / localStorage / JSON | Public metadata requests, local state and import/export | Browser APIs; no npm runtime dependency |
| Python 3 standard library | Optional local/Replit static server | PSF licence; actual team Replit execution pending |
| ReportLab; pypdf; Pillow; pdfplumber; PyMuPDF; fonttools; python-docx | Submission PDF generation and QA, not application runtime | BSD-style; BSD-3-Clause; HPND; MIT; PyMuPDF AGPL/commercial; fonttools MIT; python-docx MIT. Versions in appendix |
| Node.js and node:assert | Source checks and core logic test runner | Node.js MIT and bundled third-party notices |
| OpenAI Sites helpers; Python helper scripts | Source synchronization, packaging, persistent artifact saving | Managed OpenAI tooling under service terms |

# 3. APIs & CLOUD SERVICES

## 3.1. External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API/provider | Use | Tier |
|---|---|---|
| Arbeitnow public job-board API | GET title, tag, URL and date metadata. Europe-heavy sample. No API key. | Public free endpoint; no full-text redistribution claimed |
| HN Search / Algolia | GET AI-discussion title, URL and date metadata. Not authoritative research. | Public endpoint; publisher terms apply |
| OpenAI Sites managed tools | Development-time source, version and deployment operations. Not an app API. | Account-provided tooling; no sponsor credit claim |
| OpenAI ChatGPT / Codex | Development assistant; not called by the running application | Existing account access; not an SDC sponsor API |

# 3. APIs & CLOUD SERVICES

## 3.2. Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service/provider | Purpose | Plan/region |
|---|---|---|
| ChatGPT Sites / OpenAI; Cloudflare hosting | Public static website and managed source/version workflow | Managed account service; plan and region not exposed to builder |
| Replit | Required sponsor platform; run configuration prepared only | PENDING actual team-account use, plan, region and evidence |
| GitHub | Required public submission repository | PENDING team repository; managed Sites remote is not GitHub proof |
| RunPod | None used; no GPU or model training | N/A under conditional GPU rule; confirm sample README conflict |
| Browser localStorage | Local selections and progress, not cloud storage | User browser; clear with reset |
| Other databases/auth/storage | None in application | No database, profile auth or backend secrets |

# 4. AI TOOLS, MODELS & DATASETS

## 4.1. AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model/version | Provider | Use | Terms |
|---|---|---|---|
| None in running solution | N/A | No pre-trained, fine-tuned or self-trained model; rule-based baseline only | N/A |
| Development assistant model ID not exposed | OpenAI | ChatGPT / Codex assisted code, design and documents; see 4.3 | OpenAI service terms |

# 4. AI TOOLS, MODELS & DATASETS

## 4.2. Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset | Source/type | Rights | Personal data |
|---|---|---|---|
| Job titles/tags/dates/links | arbeitnow.com/api/job-board-api; public | Provider/publisher terms; no open-data licence asserted | No learner data; no raw descriptions retained |
| AI discussion titles/dates/links | hn.algolia.com API; public | Source/publisher terms | No profile/author fields retained |
| Fixed personas and test signals | Project-generated; synthetic | Project code Apache 2.0 subject to IP review | No |
| Role reference catalogue | O*NET® OnLine; linked public reference | CC BY 4.0 except source exceptions; modifications disclosed | No |
| Learning and SDG references | Provider pages / sdgs.un.org; links | Provider terms; content not redistributed | No |

# 4. AI TOOLS, MODELS & DATASETS

## 4.3. AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☐ No

YES - generative AI assisted development. Extent: MAJOR.

If yes, list them below:

| Tool/version | Used for | Extent |
|---|---|---|
| OpenAI ChatGPT / Codex; exact underlying model ID not exposed | Major application code and UI, logic revisions, test generation, source analysis, PDF drafts and submission guidance | Major |
| Other AI assistants / image generators | None evidenced in this workflow. Team must add any additional use. | Pending team confirmation |

# 4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

OpenAI ChatGPT / Codex generated and revised major portions of the application and submission drafts. Automated logic checks and browser checks are documented in the accompanying test report. These assistant checks do not constitute registered-team review. The team must inspect and adapt the source, validate evidence and rights, rehearse the data flow and verify all claims. Team sign-off, original problem framing and workshop completion are pending. Do not certify them until all five members can explain the core features.

# 5. SPONSOR-PROVIDED RESOURCES

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

| Sponsor/resource | Actual use | Evidence/status |
|---|---|---|
| Replit | No use evidenced in this workflow | Mandatory; team-account use and proof PENDING |
| RunPod | None; no GPU work | Applicability PENDING organizer clarification of sample README |
| Sponsor GenAI, credits, data or mentoring | None evidenced; additional team use unknown | Team must confirm and add every actual resource before signing |

# 5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

The deployed deterministic baseline needs static hosting and public metadata feeds; it does not technically require a sponsor GenAI API or GPU platform. Hosting is replaceable, and Replit run support is prepared. However, technical replaceability does not waive the competition's compulsory Replit rule. Actual Replit usage and evidence remain mandatory. RunPod would become necessary for any future GPU training, fine-tuning or inference, subject to organizer clarification of the sample README.

# 6. SECURITY, PRIVACY & RESPONSIBLE USE

## 6.1. Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

The demo permits fixed synthetic personas, skill selections, interests, weekly study hours and optional experience context. It does not accept CVs, emails or real learner identifiers. These selections, progress and public signal metadata remain in browser localStorage until reset or browser deletion. Exports are downloaded locally. Feed providers receive ordinary connection metadata but no profile fields; requests omit credentials and referrers. Hosting may keep normal service logs under provider policies, whose retention is not controlled by this app. JSON imports must be public or synthetic and pass URL/date/type validation. Raw job descriptions and HN author fields are not retained or displayed. No profile is sent to an AI service. Leader contact details appear only in submission documents, not the public app.

# 6. SECURITY, PRIVACY & RESPONSIBLE USE

## 6.2. Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

This version uses rules, not a trained forecasting model. Keyword ambiguity, small samples, English-centric vocabulary and Europe-heavy job coverage limit relevance to Malaysian learners. Within-source date groups can still be biased and are not job-loss predictions. Scenarios and study hours are design assumptions. Source outages, changed URLs and browser storage limits can affect operation. Mitigations include evidence dates, minimum sample checks, explicit score components, safe import validation, source separation and local export. Public API access does not settle all data reuse rights; verification remains pending. No empirical fairness or usability study has been completed. Replit proof, team review and AI-powered sufficiency remain eligibility issues. Human users keep final control.

# 7. COMPLIANCE & READINESS

*Tick all boxes that apply before submitting.*

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

No checklist is certified for the team. Complete repository access, sponsor evidence, rights review and registered-team approval before checking these items.

# 8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

PENDING genuine signature by LIN ZIRUI after final team review.

[Date]

PENDING actual signing date. Draft prepared: 23 October 2026.

This document records observed usage and explicitly unresolved items. It does not certify eligibility, team authorship, workshop attendance, completed sponsor use or full licence clearance.

# Requirements and verification appendix

Evidence status: review draft; unresolved fields are not certified.

## Implemented scope

Profile form with fixed synthetic personas; 9 skills; 5 role pathways; manual updates from 2 public feeds; import validation and source separation; score breakdowns and gaps; ordered introductory resources; scenario comparison; SDG mappings; local milestones and JSON export. No CV upload, runtime LLM, background monitoring, real recruitment or job guarantee.

## Acceptance and evidence

Core test command: node tests/core.test.cjs. Tests verify baseline scores, evidence effects, scenario differences, skill-gap updates, score bounds, source isolation, low-sample guards, keyword boundaries, invalid URL/date rejection, deduplication, learning order, duration and exclusion of demographic fields. Passing these checks is software verification, not a user-impact study.

## Primary references

[D1] Domain 1(1).pdf, internal pages 1-7 (PDF pages 2-8): problem, six features, rules and preliminary rubric. [O] SDC Hackathon Online Orientation Slides, PDF pp. 21-24 and 28-35: domain, AI/integrity, submissions, sponsor resources, deadline and penalties. [P] Official Phase 1 Submission Template. [T] Official Technical Tools Declaration Template.

## SDG and provider references

[UN4] https://sdgs.un.org/goals/goal4; [UN8] https://sdgs.un.org/goals/goal8. [J] https://www.arbeitnow.com/blog/job-board-api. [HN] https://hn.algolia.com/api. [ON] https://www.onetonline.org/help/license. Access checked 23 October 2026; external page access can change.

# Requirements and verification appendix

Evidence status: review draft; unresolved fields are not certified.

## Word-limit audit

Proposal 2.1: 166/250 words. Proposal 2.2: 191/300. Proposal 3.2: 171/200. Declaration 2.1: 134/200. AI review: 79/100. Privacy: 121/150. Risks: 109/150. Counts use whitespace-separated English words.

## Evaluation design and targets

Five synthetic journeys should cover the different target pathways and varied skill coverage. Record fixture ID, inputs, scenario, source snapshot, target, gap list, references, total hours and export. Baseline 0/5 is a constructed profile-only comparator. Phase 2 target >=4/5 (80%) passing journeys is proposed, not achieved. Human planning-time baseline and outcome data remain unmeasured.

## Completion blockers

PENDING: leader phone; actual team public GitHub URL; actual team-account Replit usage/proof; five members’ workshop completion; prior-work and employment-IP answers; source rights review; team understanding/adaptation of AI output; genuine leader signature/date. Runtime AI functionality is not implemented; the team should confirm whether the deterministic baseline is adequate for the domain.

## Official GitHub sample

Retrieved directly from https://github.com/michaelzhngg/SDC_Hackathon_Submission_Template using the GitHub contents/tree API on 23 October 2026. Its sample includes preliminary, technology, Replit proof and RunPod proof PDFs. The README marks RunPod required; orientation p.29 limits the RunPod rule to GPU work. Obtain written organizer clarification. Terms & Conditions take precedence; they were not provided.

# Technical tool version record

Evidence status: review draft; unresolved fields are not certified.

## Document generation only

reportlab 4.4.9; pypdf 6.10.0; Pillow 12.3.0; pdfplumber 0.11.9; PyMuPDF 1.26.6; fonttools 4.61.1; python-docx 1.2.0. Python 3.12.14. Node.js v24.19.0 used for JS checks. These are read, build and document tools, not client application dependencies. PyMuPDF rendered final QA pages; fonttools instantiated the Chinese font; python-docx inspected official templates. PDF typography: DejaVu Sans (Bitstream Vera / DejaVu licences) and Noto Sans SC (SIL Open Font License 1.1), embedded for reliable text rendering.

## Application and service versions

HTML5/CSS3; ECMAScript 2020+ features in a current browser. No npm runtime framework. Python 3 standard library server. Browser, Replit plan/region and managed hosting internals are not pinned or exposed. No runtime model version exists. Exact ChatGPT / Codex model identifier is not exposed and is not guessed.

## Sources and resource availability

Introductory resources: Kaggle Learn Python/Pandas; Google ML Crash Course; Arduino Learn; Microsoft Excel help; OpenLearn; UN CC:e-Learn. Circuit resource: All About Circuits textbook. AI ethics reference: Elements of AI. These are links, not copied course content. Search could not independently inspect the latter two sites due HTTP 403, so the team must verify their links/access manually before submission. Catalogues may contain paid certificates or optional paid products.

## Prior prototype resources

The earlier prototype included Google Fonts CSS for DM Sans and Manrope. The revised build removes all external font requests and uses Arial/system fonts. The earlier version used full-text job snippets in browser-local state; v2 stores only title/tag/date/link metadata and does not migrate old evidence. Reset removes old and new demo keys. The team should disclose any additional tools or prior sources it actually used.